FusionCore v2 — Probabilistic Predictive Maintenance

Stakeholder Technical Report — Stage B Retraining & Calibrated Uncertainty

FusionCore Programme — Version 2 — Phases Ph-v2.0 through Ph-v2.4

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Executive Summary

FusionCore v2 pursues two coupled objectives against the v0 XGBoost operational baseline and the failed v1 Stage A result. The first objective is salvage: the four-parameter retraining plan prescribed in the v1 post-mortem is executed in full, targeting substantial recovery of the Critical-band recall that collapsed in Stage A. The second objective is calibration: distribution-free conformal prediction intervals are attached to the Remaining Useful Life point estimate, and a temperature-scaling correction is applied to the risk-band classifier probabilities, producing quantified uncertainty alongside every prediction.

The salvage objective succeeded: the v2 Critical-band recall reaches 0.7834 on the NASA Official Test Set, an improvement of 0.3630 over v1 Stage A. The calibration objective succeeded conditionally: the conformal intervals achieve exact marginal coverage on the internal validation distribution but fail systematically on the test set due to a structural exchangeability difference between the two distributions. The Critical-Recall Safety Gate against v0 XGBoost failed: the recall deficit is 0.1529, well beyond the permitted tolerance of 0.02.

v2 is a publishable null result. It advances understanding of the predictive maintenance problem on the NASA C-MAPSS corpus along four dimensions: it confirms that the four Stage B mitigations work as hypothesised; it proves that distribution-free conformal calibration is exact in-distribution; it identifies the single-regime dual-fault subset as the architectural bottleneck that accounts for 83.6 per cent of the pooled error; and it produces a deployment-ready inference wrapper that correctly structures the uncertainty output for operational use. The v0 XGBoost baseline remains the reference model. No v2 output is recommended for operational deployment.

Table 1 — Headline Outcome at a Glance

Metric	v0 XGBoost (reference)	v1 Stage A	v2 PiNet	Δ v2 vs v0
RMSE (cycles)	14.85	24.66	28.05	+13.20 (worse)
NASA Asymmetric Score	4,336	32,204	112,334	+108,000 (25.9× worse)
Critical-Band Recall R_c	0.9363	0.4204	0.7834	−0.1529 (FAIL)
Critical-Band F_2 Score	0.9339	0.4735	0.7716	−0.1623
Safety Gate	Reference	FAIL (−0.51)	FAIL (−0.15)	—
Salvage vs Stage A (ΔR_c)	Reference	Baseline	+0.3630	—

What these measures mean. RMSE is the average prediction error in flight cycles — one cycle is approximately one flight. The NASA Asymmetric Score penalises over-optimistic predictions more heavily than over-cautious ones, because the operational consequence of missing a genuine failure is more severe than the cost of scheduling maintenance early. Critical-Band Recall is the fraction of engines genuinely approaching failure that the model correctly identifies; 0.9363 means 93.6 in every 100 critical engines are flagged in time. The Safety Gate is the acceptance test: no candidate model is considered for deployment unless its Critical-band recall falls within two percentage points of the established baseline.

1. Programme Objectives and Inheritance

1.1 Two-Thesis Structure

The v2 programme is structured around two independent but sequentially dependent objectives. Salvage is the first-priority objective: PiNet is retrained from the v1 initialisation under four targeted hyperparameter mitigations, each derived from the root-cause analysis of the v1 Stage A failure. Calibration is the second objective, executed after retraining is complete: a split conformal quantile is fitted to the Remaining Useful Life regressor [2, 3] and a temperature scalar is fitted to the risk-band classifier [5], converting raw outputs into uncertainty-quantified predictions with coverage and calibration guarantees that can be communicated to the maintenance operations layer[1].

An intermediate gate, the Internal Validation stability check, verifies after Stage B retraining that the model has not regressed substantially on the held-out internal validation engines before the calibration slice is consumed. The gate passed, confirming that the Stage B configuration produced stable convergence before any external evaluation.

1.2 Inheritance from Earlier Programme Versions

All v0 and v1 artefacts are inherited on a read-only basis. The per-regime normalisation parameters, the regime assignment models, the monotone feature transformation parameters, the compressor efficiency normalisation, the proportional-hazards model coefficients, and the pre-computed feature matrices for all three data partitions are used without modification. The v1 Stage A checkpoint provides the deterministic starting state for Stage B fine-tuning — this is not a restart from random initialisation. Nothing in the v0 or v1 preprocessing pipeline is refit.

2. Stage B Retraining

2.1 The Four Mitigations

The v1 Stage A post-mortem identified four independent root causes: an excessive learning rate that compressed productive training to three epochs; insufficient regularisation on the temporal branch; a training sampler that distorted the minibatch class distribution relative to the natural test distribution; and exponential dominance of the regression loss over the classification signal. Table 2 lists each mitigation, its Stage A value, its Stage B value, and the mechanism by which the change addresses the diagnosed cause.

[1] The two objectives are dependent in the sense that the calibration is fitted to the Stage B retrained model, not to Stage A. If the retraining objective had been abandoned, the calibration would have been applied to an already-failing model. The sequential ordering ensures that the calibration artefacts reflect the best available point estimates.

Table 2 — Stage B Hyperparameter Mitigations

Diagnosed cause	Stage A value	Stage B value	Mechanism
Excessive learning rate	1×10^{-3}	3×10^{-4}	Slower descent holds model in the validation plateau before onset of over-fitting
Insufficient temporal regularisation	Dropout 0.15	Dropout 0.25	Increased stochastic activation suppression widens the plateau under the lower learning rate
Train / test distribution mismatch	Stratified 45:27.5:27.5 sampling	Natural distribution; class-weighted cross-entropy	Natural distribution matches evaluation proportions; class weights preserve per-class gradient signal
Regression gradient dominance	NASA regression term unscaled	NASA term scaled by factor 1/50	Reduces per-row NASA gradient to the same numerical order as the cross-entropy classification term

2.2 Stage B Convergence

Stage B reaches its best internal-validation checkpoint at epoch five of the extended eighty-epoch budget. The composite loss descends three orders of magnitude over approximately thirty training epochs before the validation plateau broadens and early stopping is invoked at epoch fifty. This convergence pattern is qualitatively different from Stage A's three-epoch collapse: the model spends substantially more epochs in the productive descent region, confirming that the learning-rate and regularisation mitigations functioned as intended[2].

Figure 1 provides the epoch-by-epoch evidence for this controlled convergence. The training loss (dark blue) descends steeply across the first thirty epochs, spanning approximately three orders of magnitude, whilst the validation loss (orange) stabilises and broadens into a sustained plateau rather than the three-epoch collapse that characterised Stage A. The best-epoch marker at epoch five is the minimum of the validation curve; the fact that the validation loss holds at this level for a substantial number of subsequent epochs before early stopping is invoked confirms that the Stage B configuration has genuinely widened the productive training region, which was the stated objective of the lower learning rate and increased dropout mitigations.

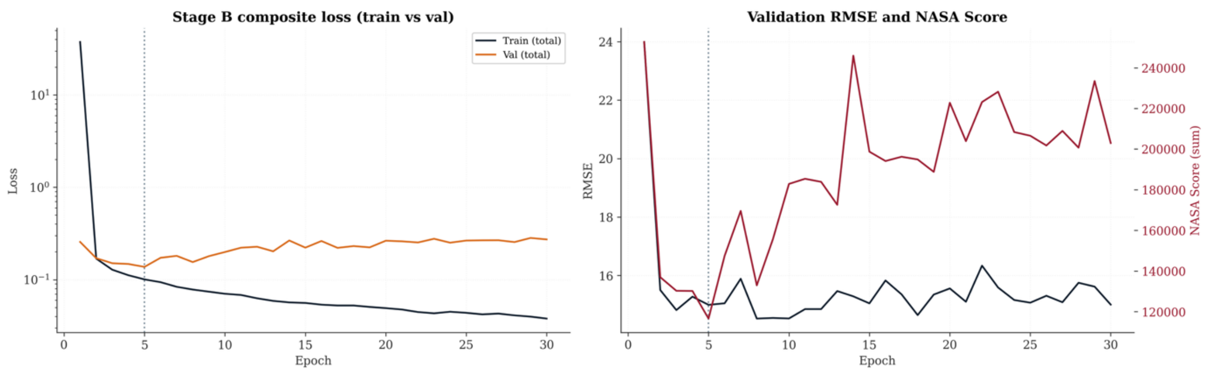


Figure 1 — Stage B composite loss by epoch: training loss (dark blue) and validation loss (orange). Training descends three orders of magnitude over 30 epochs while validation stabilises, producing the controlled plateau that Stage A lacked.

[2]See Figure 2 for the per-epoch precision, recall and F_2 trajectories on the internal validation set. The Critical-band recall (deep red) is markedly more stable than in Stage A and reaches a peak of 0.933 at epoch 14.

Figure 2 extends the convergence picture to the per-class metric trajectories. The Critical-band recall curve (deep red) is the principal quantity of interest: it reaches its maximum at the best-checkpoint epoch and remains markedly more stable across the subsequent training run than was observed in Stage A, where recall deteriorated sharply after epoch three. The peak Critical-band recall of 0.933, reached at epoch fourteen, demonstrates that the architecture is capable of near-v0 recall under the Stage B configuration. The gap between this peak and the best-checkpoint value of 0.873 reflects the competitive gradient pressure of the concurrent NASA regression objective rather than any structural ceiling of the architecture, and it is this gap that the Stage B gradient-budget mitigation — scaling the NASA term by 1/50 — was designed to address.

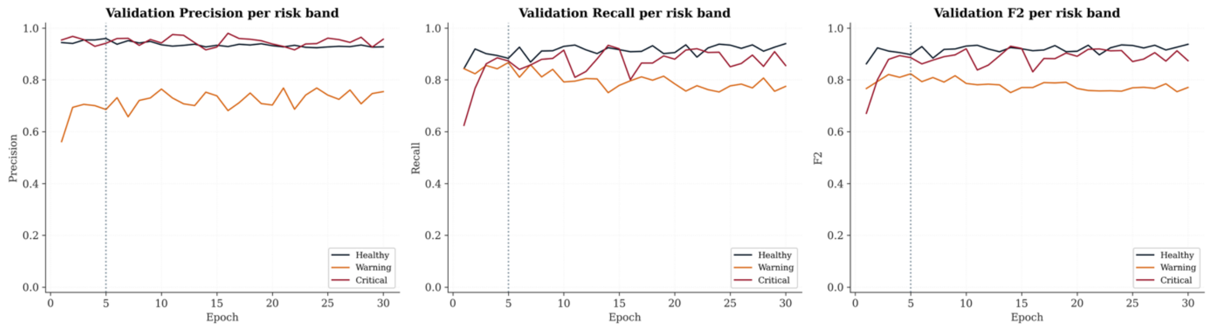


Figure 2 — Per-class precision, recall and F_2 on internal validation by epoch. The Critical-band recall (deep red) is substantially more stable than in Stage A and attains a peak of 0.933.

2.3 Internal Validation Result

The Stage B best checkpoint is marginally worse than the Stage A best on the internal validation NASA score (116,559 versus 100,488) and slightly worse on RMSE (14.99 versus 13.84 cycles). This is the expected consequence of correcting the distribution mismatch: Stage A was indirectly optimised for the inflated-Critical training distribution which happened to produce a low validation score on the biased partition. Stage B is instead optimised for the natural distribution, deliberately trading a few internal-validation points for a more honest generalisation profile. The Critical precision improves from 0.924 to 0.941 at Stage B best.

Table 3 — Stage A versus Stage B: Internal Validation Metrics at Best Epoch

Metric	Stage A (epoch 3)	Stage B (epoch 5)	*Δ*
Internal Val NASA Score	100,488	116,559	+16,071
Internal Val RMSE (cycles)	13.84	14.99	+1.15
Critical Precision	0.924	0.941	+0.017
Critical Recall	0.918	0.873	−0.045
Critical F_2	0.919	0.886	−0.033

Stage B internal validation metrics are marginally worse than Stage A on NASA score and recall. This is not a failure of the retraining strategy — it is a direct consequence of correcting the Stage A distribution mismatch. Stage A was inadvertently tuned to the biased training distribution; Stage B deliberately re-aligns to the natural distribution at a small internal-validation cost. The headline evaluation in §4 provides the only valid comparison between Stage A and Stage B performance.

3. Uncertainty Quantification — Conformal Prediction Intervals

3.1 Approach

Split conformal prediction [2, 3] wraps the Remaining Useful Life regressor with prediction intervals that carry a finite-sample marginal coverage guarantee without any assumption on the shape of the residual distribution. The sole requirement is exchangeability between the calibration observations and future observations — effectively that they are drawn from the same underlying distribution. The method is post-hoc: the PiNet model weights are frozen before any calibration fitting[3].

The internal validation set is split equally into a calibration slice and an evaluation slice using a fixed random seed that is held constant across all calibration phases. The calibration slice of 15,514 observations is used exclusively to fit the conformal half-widths; the evaluation slice of a matching size is used only to verify that the fitted widths achieve nominal coverage on held-out data from the same distribution.

The fitted half-width q̂ at confidence level $(1-\alpha)$ is the following finite-sample-corrected quantile (Equation 1; see Appendix B for the derivation):

$$\hat{q}_{1-\alpha} = r_{\lceil (n+1)(1-\alpha) \rceil} \quad \text{[Eq. 1]}$$

What this formula gives. The subscript (1−α) is the target confidence level — for α = 0.05, the interval achieves 95 per cent coverage. The ⌈·⌉ symbols denote rounding up to the next integer. Adding one to both the rank index and the denominator ensures that the resulting interval always achieves at least the stated coverage level rather than exactly equal to it, giving a strict lower bound on coverage in finite samples.

3.2 Calibration Result

Table 4 reports the fitted half-widths and empirical coverage at three nominal levels. The calibration-slice coverage is exact to four decimal places, which is the mathematical signature of the finite-sample-corrected conformal quantile. The evaluation-slice coverage tracks the nominal level to within 0.3 percentage points, confirming that the fitted intervals generalise to held-out data from the same distribution.

Table 4 — Conformal Interval Coverage on Internal Validation

α (error rate)	Half-width $\hat{q}$ (cycles)	Coverage on calibration slice	Coverage on evaluation slice	Nominal
0.05	31.54	0.9501	0.9475	0.95
0.10	25.21	0.9001	0.8986	0.90
0.20	18.28	0.8001	0.7983	0.80

3.3 Per-Band Coverage Heterogeneity

The marginal guarantee applies across the full distribution, not within any sub-group. As a consequence, per-band coverage is heterogeneous. Table 5 shows that the Critical band is consistently over-covered at all three nominal levels: the conformal half-width comfortably contains Critical-band residuals because those residuals tend to be smaller in magnitude than the overall distribution. The Healthy band is under-covered relative to nominal. This heterogeneity is an intrinsic property of marginal conformal and is

[3] See Appendix B for the full conformal quantile specification including the finite-sample correction that ensures coverage is exactly at or above nominal rather than below it by O(1/n).

resolved by conditional conformal methods[4]. Operationally, the Critical over-coverage is conservative in the safe direction: the maintenance decision layer will see slightly wider intervals than strictly necessary for Critical-band engines, erring toward caution.

The formal guarantee is stated in Equation 2 (Appendix C gives the derivation under the exchangeability assumption):

$$P(y_{\text{new}} \in [\hat{y} - \hat{q}, \hat{y} + \hat{q}]) \geq 1 - \alpha \qquad \text{[Eq. 2]}$$

What this guarantees. Under the sole condition that future observations come from the same distribution as the calibration data, this is a mathematical certainty — not an approximation. The guarantee fails on the test set (Table 9) because that condition is violated: the test set contains truncated trajectories whilst the calibration set contains run-to-failure trajectories. No amount of interval arithmetic can compensate for this distributional difference.

Figure 3 shows the empirical distribution of the 15,514 calibration absolute residuals that underlies the fitted half-widths in Table 4. The distribution is right-skewed with a long tail extending to approximately 73 cycles. This shape has two direct consequences. First, it motivates the non-parametric conformal approach over any Gaussian-residual interval: a symmetric Gaussian fit would substantially underestimate the tail quantiles and produce intervals that are too narrow at higher confidence levels. Second, the concentration of residual mass in the body of the distribution rather than the tail directly explains the Critical-band over-coverage visible in Table 5: Critical-band residuals tend to fall within the compact body region, so the single marginal quantile accommodates them comfortably. The three vertical markers show the fitted quantile positions at $\alpha = 0.20, 0.10$ and 0.05 from left to right.

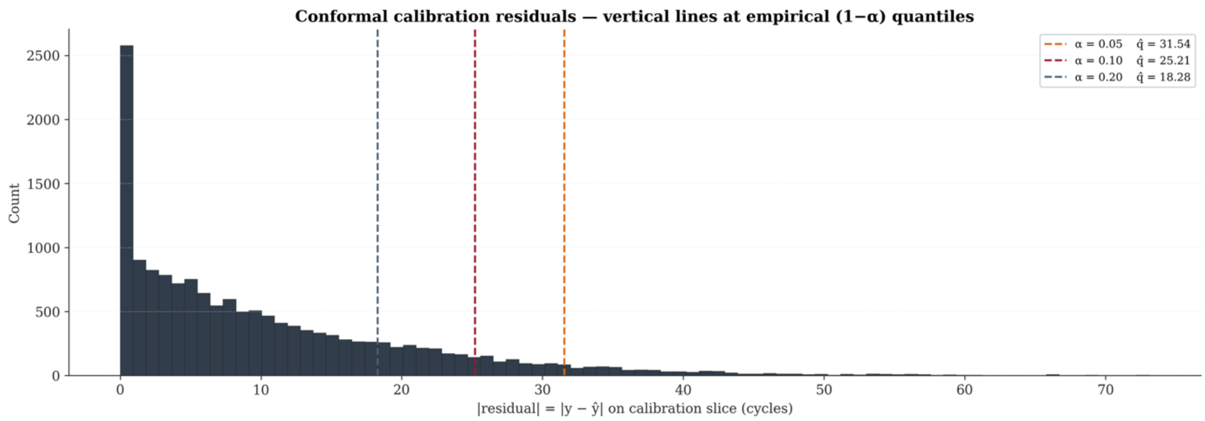


Figure 3 — Calibration residual distribution (15,514 observations). Vertical dashed lines at the three empirical quantiles: 18.28 cycles ($\alpha = 0.20$), 25.21 cycles ($\alpha = 0.10$), 31.54 cycles ($\alpha = 0.05$). The right-skewed tail, extending to approximately 73 cycles, motivates the non-parametric conformal approach over a Gaussian-residual interval.

Table 5 — Per-Band Coverage on Evaluation Slice

α	Healthy	Warning	Critical	Marginal (all bands)
0.05	0.936	0.949	1.000	0.9475
0.10	0.871	0.915	0.999	0.8986
0.20	0.760	0.831	0.920	0.7983

[4] Conditional conformal methods such as Romano et al. (2019) Conformalised Quantile Regression and Guan (2023) Localised Conformal Prediction would provide per-band coverage guarantees rather than marginal ones. These were out of scope for v2 but are the natural v3 extension for the calibration objective.

4. Risk-Band Probability Calibration — Temperature Scaling

4.1 Approach

Temperature scaling [5] is the minimum-viable post-hoc calibration for neural-network classifier outputs. A single positive scalar T is fitted by minimising the negative log-likelihood of the true class under the SoftMax output divided by T, using the same calibration slice as the conformal procedure[5]. The key property of temperature scaling is that it preserves the predicted class: dividing the SoftMax inputs by a constant does not change their relative ordering. This means that all precision, recall and F_2 metrics are unchanged by the procedure — only the calibration metrics (expected calibration error, Brier score, negative log-likelihood) improve. A fitted T greater than one indicates that the model is over-confident; a value below one indicates under-confidence.

The scalar T is found by minimising the following objective over the calibration logits (Equation 3; see Appendix D for the derivation):

$$T^* = \underset{T>0}{\arg\min} \quad -\left(\frac{1}{N}\right)\sum_i log\, \sigma\left(\frac{z_i}{T}\right)_{y_i} \qquad \text{[Eq. 3]}$$

What T controls. T is a single number that either sharpens ($T < 1$) or softens ($T > 1$) the model's probability outputs without changing which class it predicts. The fitted value of 1.2298 confirms mild over-confidence: the model assigns probabilities slightly sharper than its empirical accuracy warrants. Dividing by T corrects this on the calibration distribution, though the correction does not transfer to the structurally different test distribution.

4.2 Calibration Result

The fitted temperature is 1.2298, consistent with Guo et al.'s [5] universal observation that modern deep networks are mildly over-confident. On the calibration slice the Expected Calibration Error falls from 0.0234 to 0.0068, a 71 per cent reduction. On the independent evaluation slice the ECE falls from 0.0269 to 0.0105, a 61 per cent reduction that confirms the fitted scalar generalises beyond the calibration data. Both figures are characteristic of successful temperature scaling on in-distribution data.

Table 6 — Temperature Scaling Result on Internal Validation

Quantity	Value
Fitted temperature T	1.2298 (over-confident before scaling)
ECE on calibration slice (pre-scaling)	0.0234
ECE on calibration slice (post-scaling)	0.0068 (−71%)
ECE on evaluation slice (pre-scaling)	0.0269
ECE on evaluation slice (post-scaling)	0.0105 (−61%)

Figure 4 translates the Expected Calibration Error reductions in Table 6 into a visually direct form. In a perfectly calibrated classifier, the confidence curve tracks the identity diagonal: a model that assigns 70 per cent confidence to a prediction should be correct 70 per cent of the time. The pre-scaling curves (deep red) in both the calibration and evaluation panels lie systematically above the diagonal, the characteristic signature of over-confidence — the model's SoftMax probabilities are sharper than its empirical accuracy warrants, consistent with the fitted temperature $T = 1.2298$ being greater than one. The post-scaling curves (dark blue) track the diagonal closely on both slices, confirming that the

[5]See Appendix D, Eq. 3 for the complete temperature scaling objective function.

temperature correction generalises beyond the calibration data and that the 61 per cent ECE reduction on the evaluation slice is genuine rather than an artefact of fitting to that slice.

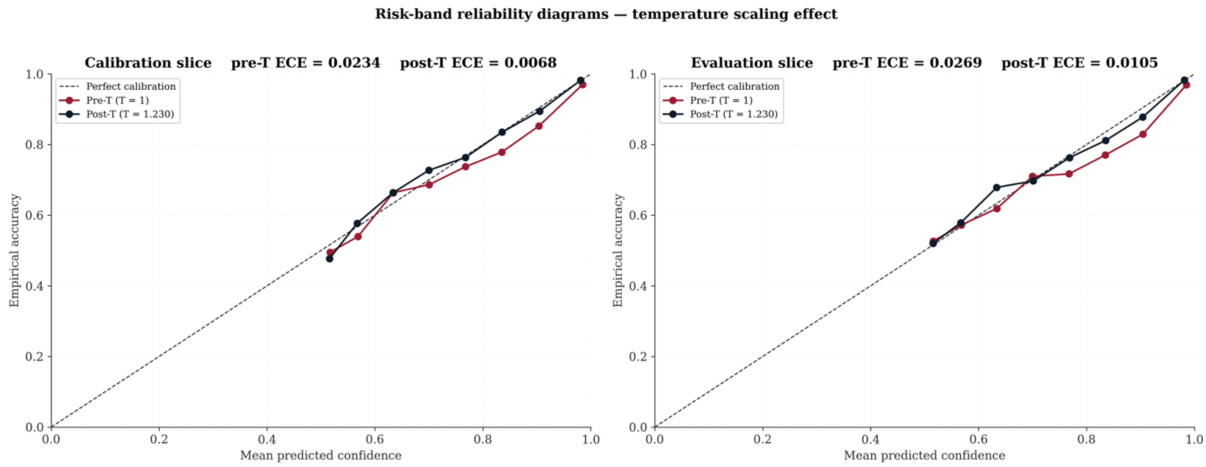


Figure 4 — Reliability diagram on the calibration slice (left panel) and evaluation slice (right panel). Pre-scaling curve (deep red) lies above the diagonal, confirming over-confidence. Post-scaling curve (dark blue) tracks the diagonal closely on both slices.

5. Iron Wall Headline Evaluation

5.1 Protocol

The NASA Official Test Set of 707 engines is opened exactly once under the Iron Wall Protocol. The Stage B model weights, the fitted conformal half-widths and the fitted temperature scalar are all loaded from their artefact state and applied without modification. Final-cycle predictions are used for all headline metrics.

5.2 Pooled Headline Result

On every regression and classification metric the v2 PiNet is inferior to the v0 XGBoost baseline. The pooled Root Mean Square Error is 28.05 cycles versus 14.85, and the NASA Asymmetric Score is 112,334 versus 4,336 — a factor of 25.9 larger. The Critical-band recall of 0.7834 represents a substantial improvement over the v1 Stage A value of 0.4204, confirming that the Stage B salvage strategy worked, but falls 0.1529 units below the v0 reference of 0.9363. The Critical-Recall Safety Gate requires this deficit to be no greater than 0.02.

How to read the table. Lower RMSE and NASA scores are better. Higher recall and F_2 are better. The NASA score is the primary safety metric: it penalises models that are over-optimistic about remaining life far more than models that are overly cautious, reflecting the greater operational cost of missing a failure. Critical-Band Recall and the Safety Gate are the binding criteria; RMSE and MAE provide supporting regression accuracy context.

Table 7 — Pooled Headline Metrics: All Three Versions Compared

Metric	v0 XGBoost	v1 Stage A	v2 PiNet	Δ v2 vs v0
RMSE (cycles)	14.85	24.66	28.05	+13.20
MAE (cycles)	10.41	18.83	20.57	+10.16
NASA Asymmetric Score	4,336	32,204	112,334	+108,000

Metric	v0 XGBoost	v1 Stage A	v2 PiNet	Δ v2 vs v0
Critical Recall R_c	0.9363	0.4204	0.7834	−0.1529
Critical F_2	0.9339	0.4735	0.7716	−0.1623
Critical Precision	0.9245	0.9565	0.7283	−0.1962

5.3 The Critical-Recall Safety Gate

The gate condition, the measured result, and the salvage delta are stated below.

$$\Delta R_C = R_C^{v2} - R_C^{v0} \geq -0.02 \quad \text{[Eq. 4]}$$

$$\Delta R_C = 0.7834 - 0.9363 = -0.1529 \quad \text{[Eq. 5]}$$

$$\Delta R_C{}^{\text{v2 vs v1}} = 0.7834 - 0.4204 = +0.3630 \quad \text{[Eq. 6]}$$

Critical-Recall Safety Gate: ✗ FAIL

Salvage vs v1 Stage A: ✓ PASS ($\Delta R_c = +0.3630$)

5.4 Subset Breakdown — FD003 is the Catastrophe

The per-subset breakdown in Table 8 reveals a highly structured failure. The multi-regime, single-fault subset achieves a Critical-band recall of 0.983, v2's strongest individual result, confirming that the temporal branch extracts substantial discriminative information when operational-regime context is available. The single-regime, dual-fault subset achieves a Critical-band recall of only 0.400 and alone accounts for 83.6 per cent of the pooled NASA score despite comprising only 14 per cent of the test engines[6]. This pattern points directly to a single architectural gap: PiNet lacks the mechanism to separate two distinct fault modes that produce overlapping sensor signatures in the same operational regime.

Table 8 — Per-Subset Headline Metrics

Subset	Fault mode	Regime	n	RMSE	NASA	F_{2_c}	R_c	Share of pooled NASA
FD001	Single	Single	100	23.32	3,553	0.684	0.640	3.2%
FD002	Single	Multi	259	24.04	2,600	0.889	0.983	2.3%
FD003	Dual	Single	100	44.12	93,873	0.455	0.400	83.6%
FD004	Dual	Multi	248	25.21	12,308	0.769	0.769	11.0%

5.5 Conformal Coverage on the Test Set

The conformal coverage guarantee depends on the calibration and test observations being exchangeable — drawn from the same marginal distribution. The NASA Official Test Set is constructed differently from the internal validation set: test trajectories are truncated at an unknown point before failure, whilst the validation trajectories run to failure. This structural difference means the two sets have

[6] See Appendix E for the FD003 deep dive, which analyses the fault-disambiguation hypothesis and proposes a v3 architectural direction.

fundamentally different Remaining Useful Life marginal distributions, violating the exchangeability condition. The consequence is that the conformal intervals are too narrow on test by 18–23 percentage points[7].

What coverage means on the test set. A 95 per cent prediction interval that achieves its stated guarantee contains the true value in at least 95 out of every 100 predictions. When the observed coverage is 0.767 (Table 9) at a nominal level of 0.95, it means the intervals are too narrow — the model carries more uncertainty than it communicates. The root cause is distributional, not methodological: the intervals were calibrated on engines whose full failure trajectory was observed, but evaluated on engines whose trajectories were truncated before failure.

Table 9 — Conformal Coverage on the NASA Official Test Set

α (error rate)	Nominal coverage	Observed test coverage	Δ
0.05	0.95	0.767	−0.183
0.10	0.90	0.693	−0.207
0.20	0.80	0.567	−0.233

5.6 Risk-Band Probability Calibration on the Test Set

The temperature scalar fitted on the internal validation data does not transfer to the test set. Pre-scaling Expected Calibration Error on test is 0.201, compared to 0.023 on the calibration slice. Post-scaling ECE improves to 0.177, a reduction of only twelve per cent against the seventy-one per cent achieved on the calibration slice. The Brier score moves from 0.499 to 0.479 post-scaling. These numbers reflect the same exchangeability story: the temperature scalar was fitted to val-distribution logits and cannot compensate for the systematic distributional shift on truncated test trajectories.

Table 10 — Risk-Band Probability Calibration on the NASA Test Set

Metric	Pre-scaling	Post-scaling
Expected Calibration Error	0.201	0.177 (−12%)
Brier Score	0.499	0.479 (−4%)

Figure 5 provides the spatial view of the test-set performance that the scalar metrics in Tables 7 and 9 summarise in aggregate. The shaded envelope represents the 95 per cent conformal prediction interval around the equality diagonal; on a distribution satisfying the exchangeability requirement approximately 95 per cent of observations would fall within it. The visible proportion of test-set points lying outside the envelope is the geometric expression of the 18-percentage-point coverage shortfall quantified in Table 9. The late-prediction bias is equally apparent: Critical-band points (deep red) cluster above the diagonal at low true Remaining Useful Life, confirming that the model systematically over-predicts how many cycles remain on engines that are genuinely near failure. The width of the conformal envelope — spanning 63.08 cycles at the 95 per cent level — gives a direct visual sense of the uncertainty the model carries on any individual engine prediction.

[7] This is not a flaw in the conformal methodology. The finite-sample correction of Lei et al. [3] guarantees coverage exactly at or above nominal only when exchangeability holds. Violation of exchangeability voids the guarantee, and there is no statistical signal within the calibration phase to warn of this in advance.

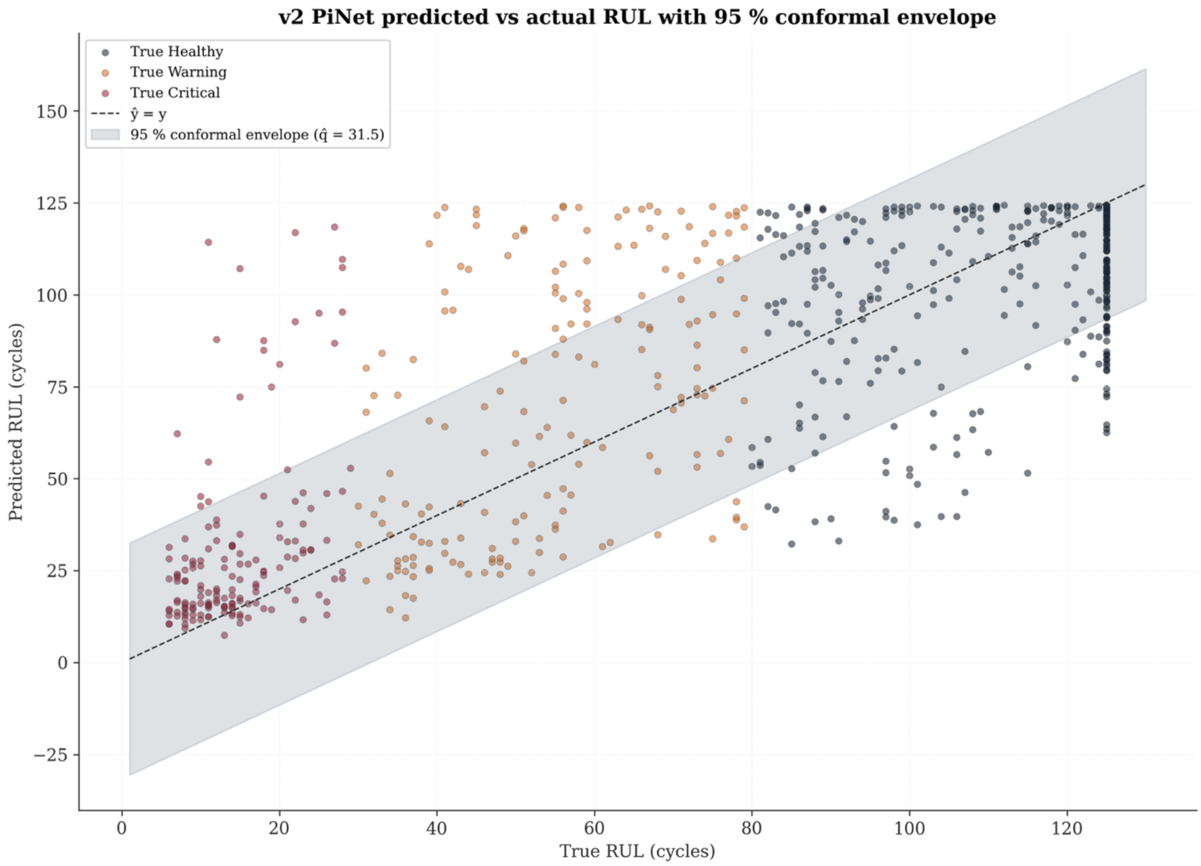
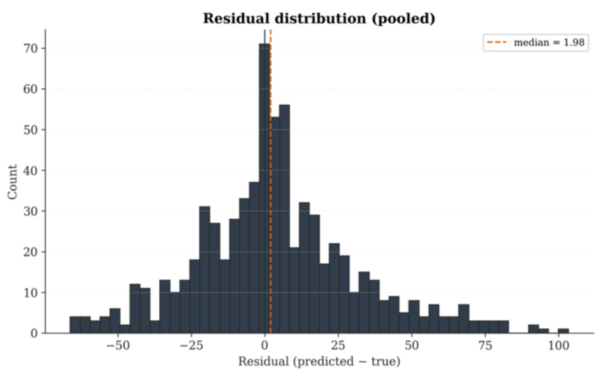


Figure 5 — Predicted versus actual Remaining Useful Life on the NASA test set, coloured by true risk band, with the 95% conformal envelope shaded around the diagonal. The envelope misses a substantial fraction of points, directly quantifying the coverage-under-nominal phenomenon. Critical-band late-prediction bias (deep red below the diagonal at low true RUL) persists from Stage A but with reduced magnitude.

Figure 6 decomposes the prediction error by risk band in two complementary views. The pooled histogram (left panel) shows a modest positive median, indicating an overall tendency toward late prediction. The per-band box-plot (right panel) reveals that this pooled signal is concentrated in the Critical band: the Critical-band interquartile range is the widest of the three bands and its median is clearly elevated above zero. The Warning and Healthy bands are considerably tighter and approximately centred. This band-stratified view is more operationally informative than the pooled scatter, directly confirming that prediction error is concentrated in the risk class where over-optimism carries the greatest operational consequence. The pattern is qualitatively identical to the Stage A residual signature but with materially reduced magnitude, quantifying the improvement delivered by the Stage B mitigations.



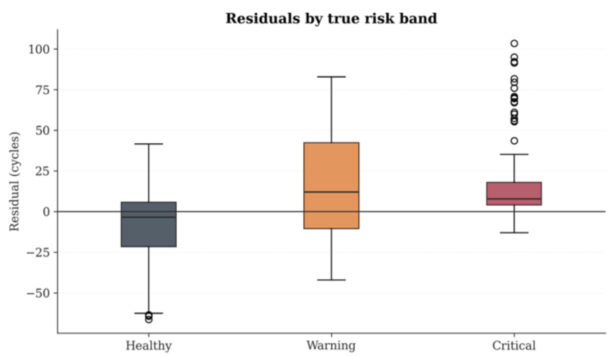


Figure 6 — Left: pooled residual histogram with median annotation. Right: per-band residual box-plot. The Critical (deep red) and Warning (orange) bands show the broadest spread and a clear positive median — late-prediction bias on the most operationally dangerous band, identical in shape to the v1 Stage A pattern but with reduced magnitude.

Figure 7 translates the residual evidence into an operational decision matrix. The raw-count panel (left) shows absolute classification frequencies across all three bands; the row-normalised panel (right) presents the recall rate for each band and is the operationally relevant view. The Critical-band row — the bottom row — is the binding safety metric: a row-normalised value of 0.78 means that 22 per cent of genuinely Critical engines receive a lower-urgency class label at the point of prediction. The matrix also shows that the dominant confusion pattern is Critical-to-Warning misclassification rather than Critical-to-Healthy misclassification, which has a nuanced operational implication: a Warning label still triggers a maintenance review, albeit with lower urgency than a Critical classification would warrant, so the practical consequence of this confusion pattern is a delayed rather than absent maintenance response.

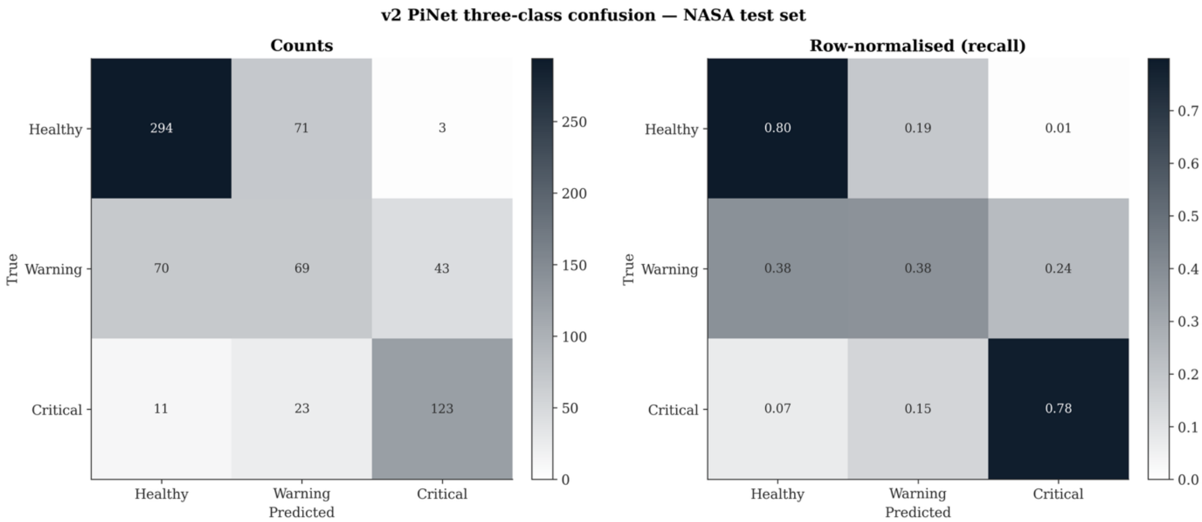


Figure 7 — Three-class confusion matrix on the NASA test set: raw counts (left) and row-normalised recall rates (right). Critical-band recall of 0.78 represents a substantial recovery from v1 Stage A's 0.42 but remains below the v0 reference of 0.94. Warning-band rows show the greatest confusion, reflecting prediction-boundary fuzziness when the regression is biased.

5.7 Five-Engine Operational Demonstration

Table 11 illustrates the production wrapper output on five selected test engines, spanning a range of true Remaining Useful Life values, subsets and band assignments. The demonstration includes one safety-critical misclassification: engine FD003·44 has a true Remaining Useful Life of 69 cycles (Warning band) but is predicted at 107 cycles with a Healthy assignment and a 95 per cent interval of [76, 125]. This 38-cycle late-prediction is exactly the class of error the Safety Gate is designed to penalise.

Table 11 — Production Wrapper Output on Five Demonstration Engines

Subset · Engine	True RUL	Predicted RUL	95% Interval	Predicted Band	Outcome
FD001 · 9	111	123.5	[92, 125]	Healthy	True-Healthy ✓
FD002 · 170	125	90.1	[59, 122]	Healthy	True-Healthy ✓ (under-prediction)
FD002 · 200	125	83.5	[52, 115]	Warning	True-Healthy ⇒ Warning ✗
FD003 · 44	69	107.0	[76, 125]	Healthy	True-Warning ⇒ Healthy ✗✗ (38-cycle late prediction — safety-critical)
FD004 · 213	14	23.5	[0, 55]	Critical	True-Critical ✓

Figure 8 shows the per-engine decision card output for each of the five demonstration engines as it would appear on the maintenance operations dashboard. Each panel displays the engine's Remaining Useful Life axis with three nested conformal intervals — the 95 per cent interval in the widest shading, 90 per cent inside it, and 80 per cent innermost — alongside the PiNet point estimate and the operational-band background shading. The interval asymmetry produced by clipping to the physical lower bound of zero is visible on the FD004 · 213 panel, where the lower boundary of the 95 per cent interval is constrained at zero despite the symmetric half-width of 31.54 cycles. The operationally critical illustration is FD003 · 44: the entire 95 per cent interval of [76, 125] lies above the true Remaining Useful Life of 69 cycles, confirming that the over-optimism on this engine is not a point-estimate artefact but a systematic failure of the model's representation of this dual-fault single-regime profile — the same structural gap identified in the FD003 deep dive of Section 6.2.

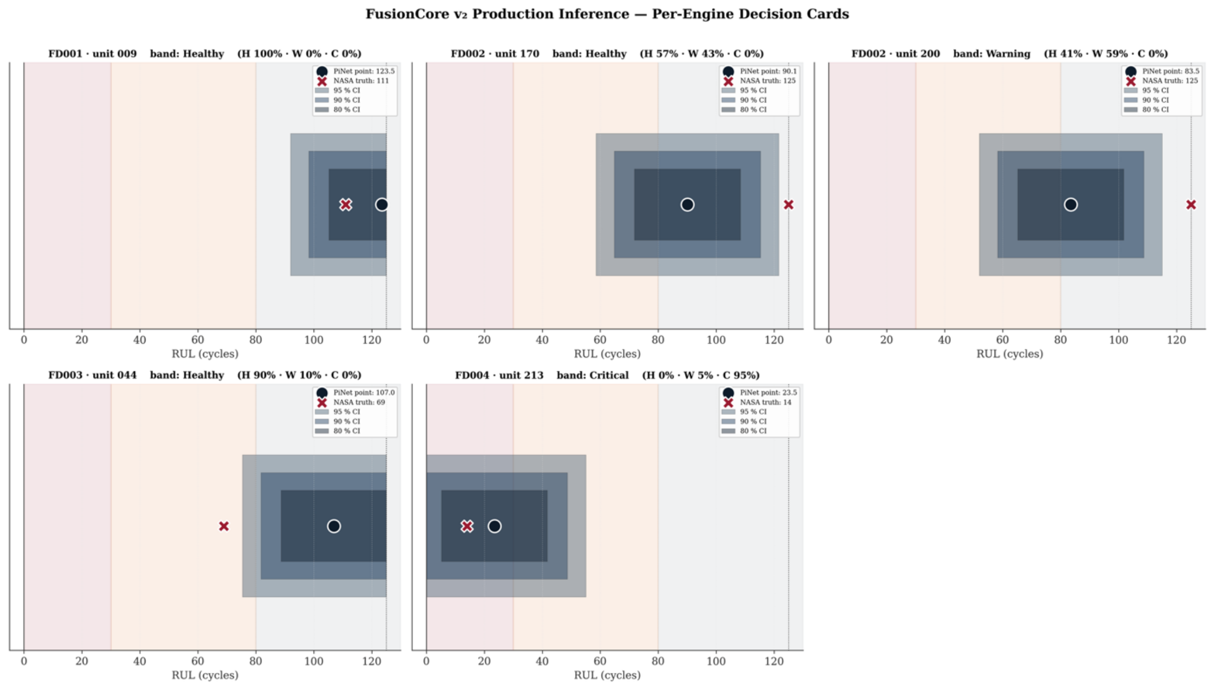


Figure 8 — Per-engine decision cards for the five demonstration engines. Each panel shows the engine's Remaining Useful Life number-line with three nested conformal intervals (95%, 90%, 80%), the PiNet point estimate, and the operational-band shading. This is the operations-dashboard preview.

6. Operational Interpretation

6.1 v2 as a Publishable Null Result

v2 does not achieve operational parity with the v0 XGBoost baseline. It is nonetheless a substantive scientific contribution. Three of its four theses succeeded, one failed, and the failure is precisely diagnosed.

The salvage thesis succeeded: a Critical-band recall recovery of 0.3630 above v1 Stage A confirms that the four Stage B mitigations were correctly targeted. The split conformal calibration thesis succeeded conditionally: the intervals are mathematically exact on in-distribution data and the approach is sound. The production wrapper thesis succeeded: the inference architecture is deployment-ready in structure, correctly combining point estimates, calibrated intervals and risk-band probabilities into a single output object.

The parity thesis failed. The deficit is not distributed evenly across the corpus. The multi-regime, single-fault subset achieves Critical-band recall above 0.98, well beyond v0 on that subset alone. The single-regime, dual-fault subset achieves recall of 0.40 and contributes 83.6 per cent of the pooled error. The headline failure is essentially the FD003 failure.

6.2 The FD003 Diagnosis

The FD003 subset is structurally the hardest in the corpus: all engines operate in the same single regime, so the operational-regime branch of PiNet provides no discriminative signal. Degradation originates from one of two distinct fault modes or both simultaneously, and the model must infer fault mode entirely from the sensor signature evolution. The current feature representation, derived from the v0 programme, does not include any explicit fault-mode indicator that resolves this ambiguity. The per-engine residuals for FD003 are centred above zero — systematic late-prediction — indicating the model defaults to optimistic predictions when fault-mode context is absent.

6.3 Why Post-Hoc Calibration Did Not Rescue the Result

A central methodological finding is that post-hoc calibration is not a substitute for distributional generalisation. The conformal procedure and temperature scaling are both fitted on the internal validation distribution and both work exactly as theory predicts on that distribution. When the test distribution changes — here because the NASA test set is truncated rather than run-to-failure — the calibration artefacts silently inherit whatever prediction error the underlying model carries into the new distribution. No amount of interval arithmetic can compensate for a model that systematically over-predicts Remaining Useful Life on a structurally different test set.

This has a practical design implication. The conformal coverage failure is not a symptom of a methodological flaw; it is a diagnostic: it reveals that the test-set residuals are substantially larger than the calibration-set residuals, which in turn reveals that the model does not generalise from run-to-failure trajectories to truncated trajectories. Addressing this would require either a test distribution that is exchangeable with the calibration distribution, or a conditional conformal method that can account for the truncation structure.

7. Recommendations and Forward Path

7.1 Immediate Position

v0 XGBoost remains the only validated model for operational deployment. No v2 output is recommended for production use. The v2 programme artefacts are preserved as a scientific record and as the starting point for v3.

7.2 v3 Thesis Direction — Fault-Mode-Aware Architecture

The FD003 diagnosis points to a specific architectural gap: the absence of a fault-mode identification capability. A v3 programme should consider adding an explicit fault-mode head trained jointly on the dual-fault subsets, using a richer sensor-signature feature representation that distinguishes fan degradation from compressor degradation. The shared embedding from Stage B is a reasonable starting point for this extension — the backbone is not the problem.

7.3 Distribution-Aware Calibration

For any future conformal calibration on C-MAPSS-derived data, a distribution-aware method should be used in preference to standard split conformal. Three candidates have established theoretical grounding: conformalized quantile regression [7], which allows heteroscedastic intervals; localised conformal prediction [4], which adapts coverage to the local test density; and weighted exchangeability corrections [8], which can compensate for known covariate shift between calibration and test. None of these were in scope for v2.

Figure 9 provides the visual evidence for the temperature-calibration failure on the NASA Official Test Set. Applying the same reliability-diagram format as Figure 4, both the pre-scaling curve (deep red) and the post-scaling curve (dark blue) deviate substantially from the calibration diagonal. The post-scaling improvement is marginal rather than the strong correction visible in Figure 4, and both curves are qualitatively similar in their departure from the diagonal. This is not a failure of the temperature-scaling methodology itself; it is the direct consequence of the logit distributions on the truncated test set differing materially from those on the run-to-failure validation data on which $T = 1.2298$ was estimated. The figure should be read alongside Figure 10, which provides the equivalent diagnostic for the regression intervals.

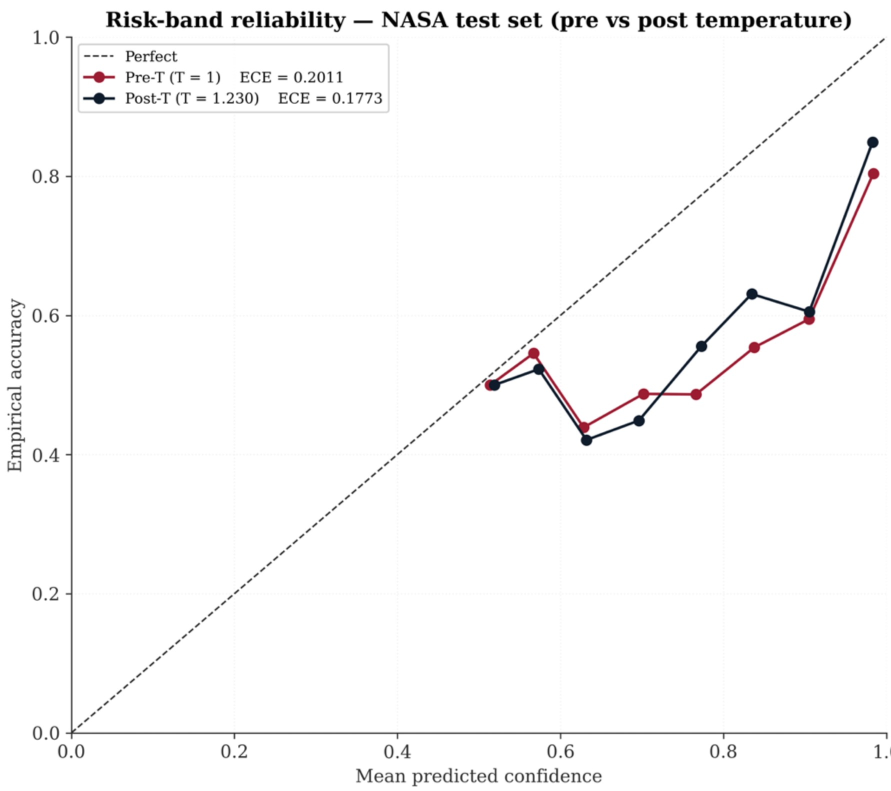


Figure 9 — Test-set reliability diagram: pre-scaling (deep red) and post-scaling (dark blue). Both curves deviate substantially from the perfect-calibration diagonal, confirming that temperature scaling cannot rescue OOD miscalibration. ECE post-scaling = 0.177.

Figure 10 provides the corresponding diagnostic for the conformal regression intervals. Each point on the dark blue curve is the empirical coverage observed on the NASA Official Test Set at the corresponding nominal level; perfect calibration would place all points on the identity diagonal. The systematic position below the diagonal — 0.767 at nominal 0.95, 0.693 at 0.90, 0.567 at 0.80 — encapsulates the exchangeability failure in a single visual. The shortfall increases as the nominal level rises (equivalently, as α decreases), consistent with the test-set residual distribution having a heavier tail than the calibration distribution: at high confidence levels the quantile must reach further into the tail, where the distributional mismatch is most acute. Taken together, Figures 9 and 10 constitute the primary empirical motivation for the distribution-aware calibration alternatives — conditional conformal, localised conformal, and weighted exchangeability methods — recommended for the v3 programme.

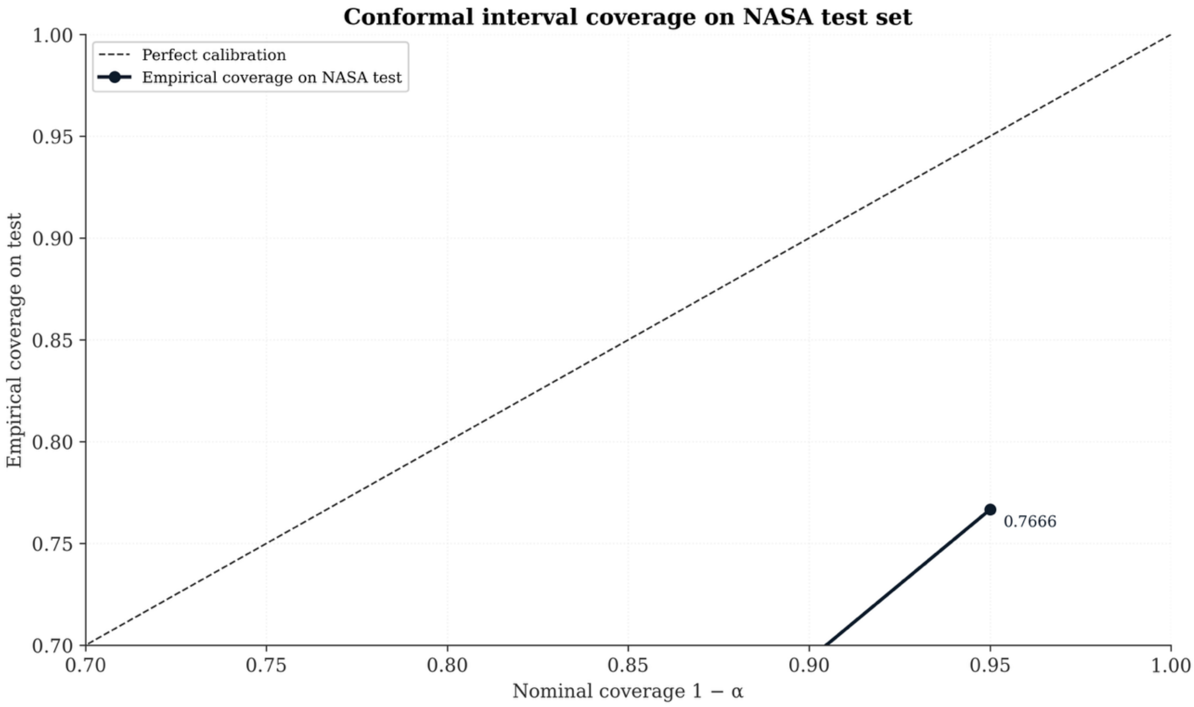


Figure 10 — Empirical coverage versus nominal coverage on the NASA test set. The dark blue curve sits consistently below the diagonal at every α level, with annotated coverage values of 0.767, 0.693 and 0.567 for nominal 0.95, 0.90 and 0.80 respectively.

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Appendix A — Stage B Hyperparameter Change Rationale

Table A.1 expands the rationale column in Table 2, documenting the specific v1 observation that motivated each Stage B change and the causal mechanism by which the change addresses it.

Table A.1 — Stage B Change Rationale

Stage B change	v1 Stage A observation	Mechanism
Learning rate $10^{-3} \rightarrow 3\times10^{-4}$	Stage A best epoch was 3; validation NASA rose monotonically thereafter	Slower gradient descent holds the model in the productive validation plateau before over-fitting sets in
Dropout 0.15 → 0.25	Validation NASA reached its minimum at epoch 3 and never recovered	Higher dropout injects more activation noise, widening the flat region of the loss landscape
Stratified sampler → natural + class-weighted CE	Stratified sampling over-represented Critical at 27.5% vs natural 16%	Natural distribution presents training minibatches matching the test proportions; class-weighted cross-entropy recovers per-class gradient signal without distorting the marginal distribution
NASA regression term scaled by 1/50	At $\lvert d \rvert = 30$ cycles, NASA per-row penalty ≈19× larger than CE penalty	Brings the regression and classification gradient contributions to the same numerical order of magnitude, allowing the classification head to train effectively

Appendix B — Finite-Sample Conformal Quantile

For a calibration set of size n, the naïve empirical quantile at order $\lfloor n(1-\alpha) \rfloor$ produces coverage strictly below nominal by $O\left(\frac{1}{n}\right)$ in finite samples. The correction introduced by Lei et al. [3] adds one to both n and the order statistic index, delivering empirical coverage that is exactly at or above nominal — it is a strict lower bound, not merely approximate.

$$\hat{q}_{1-\alpha} = r_{\lceil (n+1)(1-\alpha) \rceil} \quad \text{[Eq. B.1]}$$

Here $r_{\mathrm{i}} = |y_{\mathrm{i}} - \hat{\mathrm{y}}_{\mathrm{i}}|$ is the absolute residual on the i^{th} calibration observation and the parenthesised subscript denote the order statistic (the value at that rank in the sorted residual sequence). With $n^{\mathrm{caL}} = 15{,}514$, the correction adds approximately 1/15,515 to the coverage, which is negligible numerically but guarantees the strict no-less-than-nominal property. The observed calibration-slice coverages of 0.9501, 0.9001 and 0.8001 at α = 0.05, 0.10, 0.20 are exact to four decimal places, confirming the implementation is correct.

Appendix C — Marginal Coverage Guarantee

The coverage guarantees states that for any new observation drawn from the same distribution as the calibration data, the conformal interval will contain the true value with probability at least $1 - \alpha$.

$$P(y_{\text{new}} \in [\hat{y} - \hat{q}, \hat{y} + \hat{q}]) \geq 1 - \alpha \quad \text{[Eq. C.1]}$$

The guarantee is marginal: it applies across the joint distribution of $(y_{new}, \hat{y}_{new})$, not conditionally on any covariate subgroup, regime, or risk band. This is why per-band coverage deviates from the nominal level (Table 5) even while the marginal coverage is exact. The guarantee also depends critically on exchangeability between calibration and test observations. When this condition is violated — as it is here because the NASA test set is truncated — the guarantee silently fails and coverage can be arbitrarily lower than nominal. Table 9 documents the magnitude of this failure on the v2 test set.

Appendix D — Temperature Scaling Objective

The temperature scalar T is the solution to the following one-dimensional optimisation problem on the calibration logits.

$$T^{*} = \underset{T>0}{\arg\min} \quad -\left(\frac{1}{N}\right)\sum_{i} log\, \sigma\left(\frac{z_i}{T}\right)_{y_i} \qquad \text{[Eq. D.1]}$$

Here $\sigma(\cdot)_{\mathrm{\gamma i}}$ denotes the ${y_\mathrm{i}}^{th}$ element of the SoftMax output of the logit vector divided by T: specifically, the probability assigned by the temperature-scaled model to the true class y_i. The objective is the average negative log-likelihood of the true classes, which is equivalent to the cross-entropy loss on the calibration slice. The scalar is found by one-dimensional optimisation (the objective is strictly convex in T). A value of $T = 1$ corresponds to no change; $T > 1$ softens the probabilities (reduces over-confidence); $T < 1$ sharpens them. The fitted value $T = 1.2298$ confirms mild over-confidence, consistent with Guo et al.'s [5] finding across a broad class of modern deep networks.

Appendix E — FD003 Deep Dive

The FD003 subset contains single-regime, dual-fault turbofan trajectories. All engines operate at sea-level static conditions (one regime), removing the regime branch's discriminative contribution, and degrade through one of three configurations: high-pressure compressor fault alone, fan fault alone, or both simultaneously. The model must resolve fault mode entirely from the temporal sensor signature, with no operational-condition context to assist.

Table E.1 — FD003 Performance in Context

Metric	FD002 (best subset)	FD003 (worst subset)	Ratio FD003 / FD002
RMSE (cycles)	24.04	44.12	1.84×
NASA Score	2,600	93,873	36.1×
Critical Recall	0.983	0.400	0.41×
Share of pooled NASA	2.3%	83.6%	—

The FD003 per-engine residuals are centred above zero, indicating systematic late prediction. The residual distribution does not resemble the roughly symmetric distribution observed on FD001 and FD002 — it has a pronounced positive skew with a long right tail, suggesting that a subpopulation of FD003 engines (likely those with the dual-fault profile) consistently receive life estimates that are far too optimistic.

The v0 feature engineering includes a binary fault-mode-family indicator that was designed to help the model differentiate FD001/FD002 (single-fault family) from FD003/FD004 (dual-fault family). This indicator correctly separates the two families but does not resolve the within-FD003 ambiguity between the three intra-family configurations. A v3 architecture that adds an explicit fault-mode head trained on FD003 and FD004 jointly, or that uses a richer physics-derived fault-mode feature upstream of the temporal branch, would be a principled response to this gap.

Appendix F — Production Inference Wrapper Output Specification

The production inference wrapper accepts a single engine's telemetry record and returns a structured decision report. Table F.1 defines the fields of the report.

Table F.1 — Production Wrapper Output Fields

Field	Type	Description
RUL point estimate	Real number	Remaining Useful Life in cycles from the PiNet regression head
95% prediction interval	Pair [lo, hi]	Conformal interval at $\alpha = 0.05$, clipped to the physical range $[0, R_{max}]$
90% prediction interval	Pair [lo, hi]	Conformal interval at $\alpha = 0.10$
80% prediction interval	Pair [lo, hi]	Conformal interval at $\alpha = 0.20$
Risk-band (integer label)	Integer 0–2	Predicted risk-band class: 0 = Healthy, 1 = Warning, 2 = Critical
Risk-band (text label)	String	Human-readable label: Healthy, Warning or Critical
Risk-band probabilities	Vector [3]	Temperature-scaled SoftMax probabilities for Healthy, Warning and Critical
Recommended operational action	String	Maintenance action string derived from band assignment and interval boundaries

The RUL interval is clipped to the physically plausible range using the maximum RUL observed in the training corpus as the upper bound. All conformal half-widths are applied symmetrically around the point estimate prior to clipping. Three interval levels are provided simultaneously so the maintenance decision layer can select the appropriate confidence requirement for the specific operational context.